A Soft-Label ResNet U-Net Ensemble for Solar Filament Segmentation

A Solution to the Solar Filament Segmentation Challenge 2026

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ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. The final system averages the probability maps of two U-Nets with ImageNet-pretrained ResNet34 and ResNet50 encoders, trained at native 2048×2048 resolution on the mean of the annotators' masks (soft labels). Hysteresis thresholds, minimum area, limb erosion, and a fragment-grouping distance are tuned on 111 calibration images; all observations from one capture date stay in one data partition. On 139 held-out validation images the ensemble reaches micro Panoptic Quality (PQ) **0.4271** (date-cluster bootstrap 95% interval [0.4053, 0.4466]), compared with 0.1796 for our earlier classical baseline. For the submitted file, the same ensemble is retrained on all 707 labelled images; it scores **0.36** on the Kaggle public leaderboard, and has no held-out score of its own. We release source code, notebooks, and checksums.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [3]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [2].

Our first entry was a classical local-contrast pipeline (PQ 0.1796 on validation, 0.15 public). It misses weak filaments and confuses other dark features with filaments. We therefore replaced its pixel classifier by a supervised network and kept everything downstream unchanged: the date-grouped split, the metric, and the CSV format. This report describes that network system. The classical baseline remains in the repository for comparison.

2 METHODOLOGY

2.1 Data, supervision, and external weights

The archive contains 707 training images, 180 test images, and one annotation file with 1,154 annotator–image records (8,199 filament polygons). A source image has one, two, or three independent annotation sets (411, 145, and 151 images). Only the supplied segmentation polygons are used as supervision. Chirality labels, spines, bounding boxes, external images, and test labels are not used. The only external resource is the set of publicly available ImageNet-pretrained `torchvision` weights for ResNet34 and ResNet50 [4, 6], free for every participant. The first convolution is reduced to one input channel by summing its three input-channel kernels.

2.2 Data partitions and protocol

Capture dates are assigned to partitions by a SHA-256 ordering within each year, so every image and annotator set from a date stays together. This gives 568 development images, of which 111 form the calibration subset, and 139 validation images. In every experiment the networks train on the 457 development images outside the calibration subset; checkpoints, post-processing, and the choice among candidates use calibration only, and validation is scored once per candidate. The final submission is the only exception: its networks are retrained on all 707 labelled images with settings fixed in advance (Section 2.5).

2.3 Network and training

The network (`FilamentUNet`) is a U-Net [7] whose encoder is a pretrained ResNet [4]. Because the ResNet stem halves the resolution at once and thin barbs are only a few pixels wide, a shallow two-layer branch at full input resolution is concatenated with the upsampled decoder output before a small prediction head. Input images are standardized with the mean and standard deviation of the solar disk.

The training target of an image is the mean of its annotators' masks. The metric scores a prediction against each annotator set separately, so this consensus is the best single probability map to aim for. The loss is binary cross-entropy plus soft Dice, both accepting soft targets. Training uses random 768×768 crops (70% centred near an annotated pixel), dihedral flips, brightness and contrast jitter, and Gaussian noise. We use AdamW [5] with learning rate $5 \cdot 10^{-4}$, weight decay 10^{-4} and a one-cycle schedule [8], 32 epochs of 150 steps with batch size 8, mixed precision, and an exponential moving average of the weights (decay 0.998) that is used for evaluation and inference. The ResNet34 and ResNet50 models use seeds 2027 and 2028. Inference runs on the full image and averages four flip views.

2.4 Post-processing and instances

The probability map is thresholded with hysteresis: pixels above a low threshold form regions, and a region is kept only if its peak reaches a high threshold. The solar disk is estimated per image by Otsu thresholding and eroded by 10 pixels to remove limb artifacts. Regions smaller than a minimum area are dropped. Fragments within about eight pixels of each other are merged into one filament, because ground-truth filaments are single polygons (none of the 8,199 contains several), while predictions often break up. Merging

[1] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

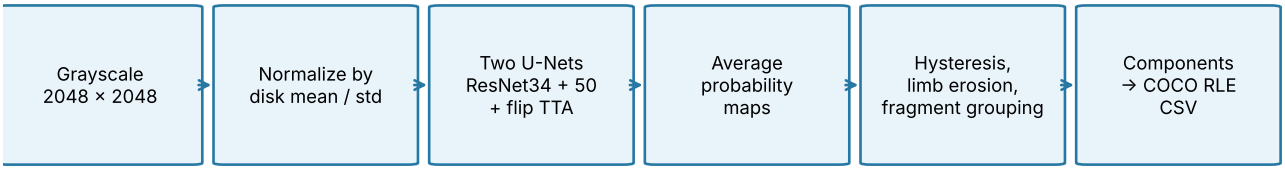


Figure 1: Inference pipeline. Each test image is processed at native resolution by both networks with four flip views; the mean probability map is turned into instances by the post-processing of Section 2.4.

only labels fragments together; it adds no pixels. Each instance becomes one COCO run-length encoded row. The values were chosen by coordinate descent on calibration PQ over the grid in the repository: low threshold 0.6, high 0.9, minimum area 300, limb erosion 10, grouping distance 4 (hole filling off).

2.5 Candidates, selection, and the refit

We compared five candidates (Table 1) and selected the two-network ensemble because it had the highest calibration PQ (0.4240). The ResNet50 alone scored slightly higher on validation, but we did not choose by validation, because that would leak. For the submitted file we retrained the two networks, with identical settings and seeds, on training plus validation (707 images), kept the final EMA weights because no held-out images remain for checkpoint picking, and reused the post-processing tuned on calibration for the ensemble trained on 457 images. No parameter was re-tuned.

3 EVALUATION

3.1 Metric

Ground-truth polygons are rasterized with `pycocotools`. For each annotator–image record, a prediction and a ground-truth instance match when IoU > 0.5; unmatched predictions are false positives and unmatched ground truths false negatives. Counts and matched IoUs are pooled over records, following the official self-evaluation notebook [1]:

$$\mathrm{PQ} = \frac{\sum_{(g,p):\mathrm{IoU}(g,p)>0.5} \mathrm{IoU}(g,p)}{\mathrm{TP} + 0.5\mathrm{FP} + 0.5\mathrm{FN}}. \tag{1}$$

Uncertainty uses 2,000 bootstrap resamples of capture-date clusters (seed 2026). The interval reflects variation within this validation partition, not seed variance or unseen-date shift.

3.2 Results

All network candidates beat the baseline by a large margin, and their intervals overlap (Figure 2), so the differences among them are small relative to the uncertainty. Instance grouping adds about 0.005 PQ. Two runs of the same v1 configuration differed by about 0.005 PQ (0.4060 and 0.4111) because GPU kernels are not bit-deterministic.

Public scores (Table 2) of the networks are about 0.08 below their local validation PQ, so we expect the private score to be lower than 0.4271 as well. The refit has no held-out score; on the test images it agrees closely with the 457-image ensemble (matched PQ 0.884 between the two files, mean pixel Dice 0.956), and its public score is 0.01 higher. Kaggle shows only two decimals and the test set has 180 images, so this does not prove that the extra data helps.

Table 1: Validation micro PQ (139 images, 222 annotator sets) of candidates trained on 457 images. "Cal." is the calibration PQ used for selection.

Candidate	Cal.	Val. PQ	TP/FP/FN
Classical baseline	0.176	0.1796	444/1160/1114
v1 U-Net, ResNet34, 512 px	–	0.4111	1047/813/511
ResNet34, 768 px, EMA	0.417	0.4208	1015/674/543
ResNet50, 768 px, EMA	0.420	0.4295	1020/644/538
Ensemble R34 + R50	**0.424**	0.4271	1033/689/525

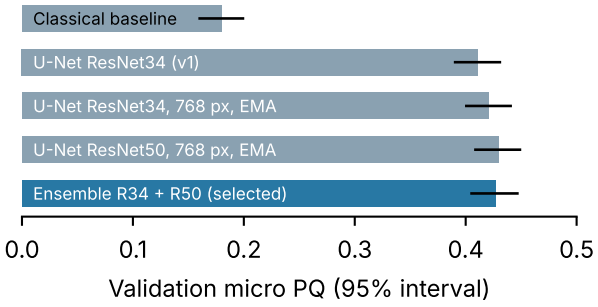


Figure 2: Validation PQ with 95% date-cluster bootstrap intervals.

Table 2: Kaggle public leaderboard scores of the files we submitted for evaluation.

Submission	Public score
Classical baseline	0.15
v1 U-Net	0.33
Ensemble trained on 457 images	0.35
Ensemble refit on 707 images (final file)	**0.36**

3.3 Error analysis and limitations

Annotators disagree strongly under this metric: scoring one annotator's filaments against another's gives PQ 0.353 on the 101 images with two or more annotators. Our predictions exceed that because they target the annotators' consensus. The consensus mask itself reaches PQ 0.78 on calibration, which is an optimistic upper reference because single-annotator images are scored against their own mask. On validation, false positives (689) outnumber false negatives (525). The system is selected on one split, one seed per network and one calibration subset. Models are trained on a single archive of GONG images, so other instruments or epochs are untested. A larger grouping distance hurts: the consensus mask drops from PQ 0.78 to 0.68 at 32 pixels.

3.4 Reproduction

Code is in the public repository; `notebooks/gpu-experiments-v2.ipynb` reproduces the candidate comparison and `notebooks/gpu-refit-707.ipynb` the final file on a Kaggle T4×2 (about 2.8 h and 1.4 h). The file submitted to Kaggle is `refit/submission.csv` (1,459 rows, 180 test images, 178 with predictions; SHA-256 begins `0fe19db5f1614a6a`; the full digest is in the repository), verified for column format, unique identifiers, and canonical RLE. Trained weights are not stored in git; they are recreated by the notebooks, and the refit notebook reads the models and post-processing from the recorded candidate-comparison run. The classical code is unchanged. AI assistance was used to develop and document the code; the participant must independently understand and defend its operation under the competition rules.

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REFERENCES

[1] Azim Ahmadzadeh. 2026. Self Evaluation Notebook. https://www.kaggle.com/code/azimahmadzadeh/self-evaluation-notebook. Official competition evaluation reference.

[2] Azim Ahmadzadeh, Rohan Adhyapak, Kartik Chaurasiya, Laxmi Alekhya Nagubandi, V. Aparna, Petrus C. Martens, Alexei Pevtsov, Luca Bertello, Alexander Pevtsov, Naomi Douglas, Samuel McDonald, Apaar Bawa, Eugene Kang, Riley Wu, Dustin J. Kempton, Aya Abdelkarem, Patrick M. Copeland, and Sri Harsha Seelamneni. 2024. A dataset of manually annotated filaments from H-alpha observations. In *Nature Scientific Data (Nature Scientific Data, Vol. 11)*. Springer, Article 1031, 1031 pages. https://doi.org/10.1038/s41597-024-03876-y

[3] Azim Ahmadzadeh, Dustin J. Kempton, Qin Li, and Alexei A. Pevtsov. 2026. Solar Filament Segmentation Challenge 2026. https://kaggle.com/competitions/filament-segmentation-2026. Kaggle.

[4] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

[5] Ilya Loshchilov and Frank Hutter. 2019. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations (ICLR)*.

[6] PyTorch Contributors. 2026. torchvision: PyTorch's computer vision library, ImageNet-pretrained ResNet34 and ResNet50 weights. https://pytorch.org/vision/.

[7] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.

[8] Leslie N. Smith and Nicholay Topin. 2017. Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates. *arXiv preprint arXiv:1708.07120* (2017).